

The Angle

The Riemann Hypothesis as a Divergence Condition

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Independent

Abstract

Let θ_N be the angle between the residual $r_{N-1} = 1_{(0,1)} - f_{N-1}^*$ and the N -th Beurling–Nyman basis direction v_N . We prove:

$$d_N = d_1 \prod_{j=2}^N |\sin \theta_j|,$$

and therefore

$$d_N \rightarrow 0 \iff \sum_{j=1}^{\infty} \cos^2 \theta_j = \infty.$$

The Riemann Hypothesis is equivalent to the divergence of $\sum \cos^2 \theta_j$: the angles between the residual and successive basis directions must not converge to $\pi/2$ too fast. This is a spectral gap condition. The condition is sharp: $\cos^2 \theta_j \geq C/j$ for some $C > 0$ would imply RH.

1 The Angle Sequence

Definition 1 (The angle sequence). At step N , the residual is $r_{N-1} = 1_{(0,1)} - f_{N-1}^*$, and the new basis direction is v_N (the component of $\rho_{1/N}$ orthogonal to $\mathcal{V}_{N-1}$). Define the *alignment angle*:

$$\theta_N = \angle(r_{N-1}, v_N) \in [0, \frac{\pi}{2}],$$

i.e., $\cos \theta_N = \frac{|\langle r_{N-1}, v_N \rangle|}{\|r_{N-1}\| \|v_N\|}$. The *alignment* at step N is $\cos^2 \theta_N \in [0, 1]$.

Theorem 2 (Product formula for the entropy).

$$d_N^2 = d_1^2 \prod_{j=2}^N \sin^2 \theta_j.$$

Proof. The decrement formula [1]:

$$d_N^2 = d_{N-1}^2 - \frac{|\langle r_{N-1}, v_N \rangle|^2}{\|v_N\|^2} = d_{N-1}^2 \left(1 - \frac{|\langle r_{N-1}, v_N \rangle|^2}{\|r_{N-1}\|^2 \|v_N\|^2} \right) = d_{N-1}^2 \sin^2 \theta_N.$$

Iterating from $N = 2$ to N : $d_N^2 = d_1^2 \prod_{j=2}^N \sin^2 \theta_j$. □

Corollary 3. $d_N = d_1 \prod_{j=2}^N |\sin \theta_j|$. The entropy at step N is the initial entropy times the product of all alignment factors $|\sin \theta_j| \leq 1$. Each step can only reduce the entropy. The product converges to 0 or to a positive limit.

2 The Divergence Criterion

Theorem 4 (RH as divergence of $\sum \cos^2 \theta_j$).

$$d_N \rightarrow 0 \iff \prod_{j=2}^{\infty} \sin^2 \theta_j = 0 \iff \sum_{j=2}^{\infty} \cos^2 \theta_j = \infty.$$

Therefore:

$$\text{RH} \iff \sum_{j=1}^{\infty} \cos^2 \theta_j = \infty.$$

Proof. The product $\prod_{j=2}^{\infty} (1 - a_j)$ with $0 \leq a_j \leq 1$ satisfies:

$$\prod_{j=2}^{\infty} (1 - a_j) = 0 \iff \sum_{j=2}^{\infty} a_j = \infty.$$

(Standard result: $\prod (1 - a_j) > 0$ iff $\sum a_j < \infty$, for $a_j \in [0, 1)$.) Setting $a_j = \cos^2 \theta_j = 1 - \sin^2 \theta_j$: $\prod \sin^2 \theta_j = 0$ iff $\sum \cos^2 \theta_j = \infty$.

By Theorem 2: $\lim_N d_N^2 = d_1^2 \prod_{j=2}^{\infty} \sin^2 \theta_j$.

If $\prod \sin^2 \theta_j = 0$: $d_N^2 \rightarrow 0$, so $d_N \rightarrow 0$. If $\prod \sin^2 \theta_j = c > 0$: $d_N^2 \rightarrow d_1^2 c > 0$, so $d_N \not\rightarrow 0$.

By the Beurling–Nyman criterion: $d_N \rightarrow 0$ iff RH. $\square$

Remark 5 (Geometric interpretation). $\cos^2 \theta_N$ measures how well the new direction v_N aligns with the current residual r_{N-1} . $\cos^2 \theta_N = 1$: perfect alignment — v_N points exactly at the residual. $\cos^2 \theta_N = 0$: perpendicularity — v_N is orthogonal to the residual, adds no information, contributes nothing to error reduction. $\sum \cos^2 \theta_N = \infty$: the new directions never become *asymptotically orthogonal* to the residual. The residual remains “visible” to the basis. RH says the zeros are all on the critical line: the error is never invisible to the Beurling–Nyman basis.

3 The Spectral Gap Condition

Theorem 6 (Sufficient condition for RH). *If there exists $C > 0$ such that*

$$\cos^2 \theta_N \geq \frac{C}{N} \quad \text{for all sufficiently large } N,$$

then $\sum \cos^2 \theta_N = \infty$ and RH holds.

More generally, any condition implying $\sum \cos^2 \theta_N = \infty$ proves RH. Examples:

1. $\cos^2 \theta_N \geq C/N$ (borderline).

2. $\cos^2 \theta_N \geq C/N^\alpha$ for $\alpha \leq 1$.
3. $\cos^2 \theta_N \geq C/\log N$ (very slow decay).
4. $\limsup \cos^2 \theta_N > 0$ (bounded below).

The condition $\cos^2 \theta_N \geq C/N^{1+\epsilon}$ for $\epsilon > 0$ does NOT suffice ($\sum N^{-1-\epsilon} < \infty$).

Theorem 7 (Necessary condition for RH failure). *If RH fails — if $\zeta(\rho_0) = 0$ with $\operatorname{Re}(\rho_0) > \frac{1}{2}$ — then $d_N \geq c_{\rho_0} > 0$ for all N , and therefore $\prod \sin^2 \theta_j \geq c_{\rho_0}^2/d_1^2 > 0$, and therefore $\sum \cos^2 \theta_j < \infty$.*

Equivalently: if $\sum \cos^2 \theta_j = \infty$, then RH holds.

The contrapositive: $\sum \cos^2 \theta_j < \infty$ implies RH fails. So ANY convergence of $\sum \cos^2 \theta_j$ would disprove RH (and would give an explicit off-line zero via the orthogonal complement).

Remark 8 (What controls the angle). The alignment $\cos^2 \theta_N = |\langle r_{N-1}, v_N \rangle|^2 / (d_{N-1}^2 \|v_N\|^2)$ depends on:

1. The residual r_{N-1} : the part of $1_{(0,1)}$ not yet approximated. This encodes the arithmetic not yet captured in $\mathcal{V}_{N-1}$.
2. The new direction v_N : the part of $\rho_{1/N}$ not already in $\mathcal{V}_{N-1}$. This depends on the Gram determinant $\det G_N / \det G_{N-1}$.
3. Their inner product $\langle r_{N-1}, v_N \rangle = \langle r_{N-1}, \rho_{1/N} \rangle$: how much does $\rho_{1/N}$ correlate with the remaining error? Via the Mellin transform (see [2]): this inner product is controlled by $\zeta(1/2 + it)$ oscillated at frequency $\log N$. As $N \rightarrow \infty$, the oscillation becomes rapid (Riemann–Lebesgue), and $\cos^2 \theta_N \rightarrow 0$. The question is how fast.

4 Connection to the Gram Matrix

Proposition 9 (Angle in terms of Gram determinants).

$$\cos^2 \theta_N = \frac{|\langle r_{N-1}, \rho_{1/N} \rangle|^2}{d_{N-1}^2} \cdot \frac{\det G_N / \det G_{N-1}}{\det G_N / \det G_{N-1}} = \frac{|\langle r_{N-1}, \rho_{1/N} \rangle|^2}{d_{N-1}^2 \cdot (\det G_N / \det G_{N-1})}.$$

Equivalently:

$$d_N^2 - d_{N-1}^2 \text{ [wait, this is wrong sign]}$$

Correctly: the decrement is $d_{N-1}^2 - d_N^2 = d_{N-1}^2 \cos^2 \theta_N$, so

$$\cos^2 \theta_N = \frac{d_{N-1}^2 - d_N^2}{d_{N-1}^2}.$$

And from the decrement formula:

$$\cos^2 \theta_N = \frac{|\langle r_{N-1}, \rho_{1/N} \rangle|^2}{d_{N-1}^2 \|v_N\|^2} = \frac{|\langle r_{N-1}, \rho_{1/N} \rangle|^2}{d_{N-1}^2 \cdot (\det G_N / \det G_{N-1})}.$$

Remark 10 (Gram matrix lower bound implies RH). If $\det G_N / \det G_{N-1} \leq A/N$ (the new direction has squared norm $\leq A/N$, consistent with $\|\rho_{1/N}\|^2 \sim A/N$ from [2]), and if $|\langle r_{N-1}, \rho_{1/N} \rangle| \geq B/N$ for some $B > 0$ whenever $d_{N-1} > 0$, then:

$$\cos^2 \theta_N \geq \frac{B^2/N^2}{d_{N-1}^2 \cdot A/N} = \frac{B^2}{A \cdot d_{N-1}^2 \cdot N}.$$

For this to imply $\sum \cos^2 \theta_N = \infty$: if $d_{N-1} \geq c > 0$ (i.e., RH fails), then $\cos^2 \theta_N \geq B^2/(Ac^2N)$ and $\sum \cos^2 \theta_N = \infty$, contradicting $d_N \rightarrow 0$. This is the contradiction: the assumption that RH fails forces the alignment sum to diverge, which forces $d_N \rightarrow 0$, which contradicts the assumption.

The gap: proving $|\langle r_{N-1}, \rho_{1/N} \rangle| \geq B/N$ when $d_{N-1} \geq c > 0$ is equivalent to proving RH.

5 The Spectral Gap Statement

Conjecture 11 (Angle lower bound). *There exists $C > 0$ such that*

$$|\langle r_{N-1}, \rho_{1/N} \rangle| \geq \frac{C\|v_N\|}{N^{1/2}}$$

whenever $d_{N-1} \geq \epsilon > 0$. This would give $\cos^2 \theta_N \geq C^2/N$ and $\sum \cos^2 \theta_N = \infty$, proving RH.

Remark 12 (The angle is archimedean). The alignment $\cos^2 \theta_N$ is controlled by the inner product $\langle r_{N-1}, \rho_{1/N} \rangle$. Via the Mellin transform on the critical line (see [2]): this inner product involves $|\zeta(1/2 + it)|^2$ modulated by $(N)^{-1/2-it}$. The $(N)^{-1/2-it} = N^{-1/2}e^{-it \log N}$ oscillates at frequency $\log N$. The alignment measures whether the residual's Mellin transform has a component at the frequency $\log N$ in $|\zeta|^2$. As N grows, the frequency increases — the alignment is a statement about the high-frequency behavior of r_{N-1} relative to $|\zeta|^2$ on the critical line. The primes enter through the oscillation of ζ on the critical line. The Wall is archimedean: it lives at $\log N \rightarrow \infty$.

Theorem 13 (Summary). *Let θ_N be the angle between the N -th residual and the N -th basis direction. The following are equivalent:*

1. *The Riemann Hypothesis.*
2. *$d_N \rightarrow 0$ (Beurling–Nyman criterion).*
3. *$\prod_{N=2}^{\infty} \sin^2 \theta_N = 0$.*
4. *$\sum_{N=2}^{\infty} \cos^2 \theta_N = \infty$.*
5. *The alignment sequence $\{\cos^2 \theta_N\}$ is not in ℓ^1 .*
6. *$1/\zeta(s)$ lies in the L^2 -closure of Dirichlet polynomials $\{P_N(s) = \sum_{k=1}^N c_k(k^{-s} - k^{-1})\}$ on $\text{Re}(s) = \frac{1}{2}$ with weight $(1/4 + t^2)^{-1}$ ([2]).*

The angle sequence is a clean proxy for all of these: a non- ℓ^1 angle sequence = RH. The Wall is the question: is $\{\cos^2 \theta_N\}$ in ℓ^1 ? The primes determine the angles. The angles determine the zeros. The zeros determine the angles. The fold has always been there.

References

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